← خطوة ٤ : بنحط فوق الcollecting ← 70% etha-

← خطوة ٥ : بنقلهم لل spin colomn

← باقي الخطوات : * washing buffer → RWI → Guanidine thio-
↳ RPE

* Elution buffer → RNase free water
↳ It is unfavorable to surface binding high pH and low ionic strength.

- mRNA have (A tail) at 3'end (polyA tail)
- Oligo (dT) probes can be used to purify mRNA from RNAs
- mRNA eluted from oligo (dT) by water or low-salt buffer

لما يرتبط ال (polyA tail) مع ال oligo :
↲ rRNA & tRNA washed away
↲ purified mRNA eluted in water or low-salt buffer.

Problems of RNA isolation: RNases

Solution

* RNASes Control :-

1. Autoclave → getting rid of DNases and RNases

2. 0.1% DEPC

3. have dedicated micropipettes or use aerosol barrier tips

4. wear glove → because skin cells also produce RNases

5. Bake glass, metal, ceramic equipment at high temp

lab 4

*Plasmid isolation from E.coli using spin/selica/adsorption colomn.

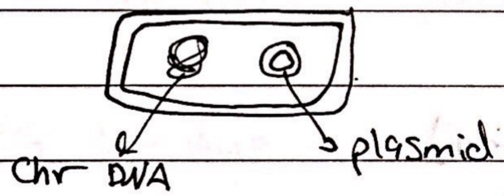
- plasmid: extra chromosomal DNA
  (small, double stranded, circular pieces of DNA)

main shape → super coiled

- plasmid replication → independently (self-replication)
  → within host genome (episomes)

- plasmids have few genes (less than 30) → not necessory for growth.

- not all of bacteria has plasmid.



* types of plasmids :

1- Virulence plasmid : ex: TI
   ↳ pharnogenic

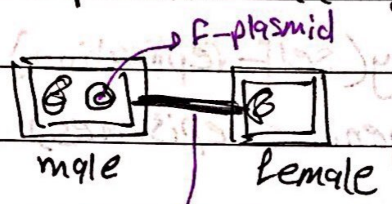
2- R-plasmid : (resistance) protect from environmental factor, MDR (multiple drug resistance)

3- Col-plasmid : for protiens that kill other microbes

Scanned with CamScanner

4- Degradative plasmid
contain digestive enzymes that allow host to digest uncommon substances (ex: toluene / salicylic acid)

5- F-plasmid (conjugative):



* بعد ما ينتقل ال F-plasmid من ال male لل female بيصير البلازميد independent عن ال chr.DNA بيصير male

* بيصير دوامج بوصل وبتكون channel بينهم بتنقل فيها ال genetic material ما عدا ال chr DNA لانه كبير الحجم.

6- HFr: (genomic recombination)
نفس ال F-plasmid بس الفرق:
لما ينتقل البلازميد بيكون عنده خيارين
إما يضل independant
او يعمل integrate لل chr. DNA

* لما نحكي tangled DNA يعني ← ال DNA ما بنقدر نرجعها لطبيعتها

*

nucleas free water	RNas free water
فش فيها RNase	فش فيها RNAas
DNAase	

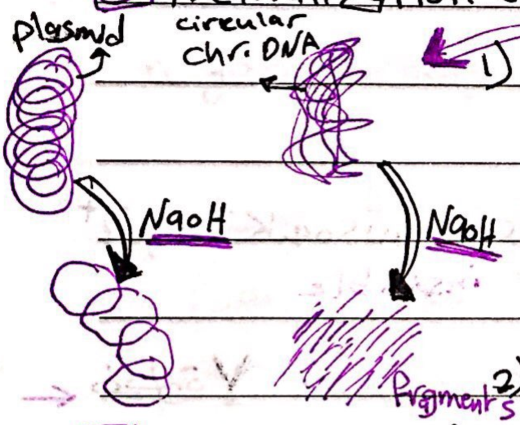
ما بصير نبدلهم ببعض

* Alkalin adsorption method :

[1] bacteria pellting → EDTA
→ Tris

[2] bacteria lysis → SDS
→ NaOH : denature Chr.DNA & plasmid

[3] Neutralization of NaOH → Potassium acetate : pH ~5



1) renatures plasmid DNA quickly while high
but Chr.DNA become tangled due to
its complexity & precipitate.
plasmid remains in solution.

2) converts soluble SDS to insoluble PDS

[4] Adsorption of plasmid → Sodium Salt : cation bridge
between silica & plasmid

[5] Washing → ethanol
→ Salt

[6] Elution → RNASe free water : remove choatropic Salt
and Sodium salt

* procedure :

cell lysis :
- SDS : $CH_3CH_2C_{11}SO_4Na$
- NaoH : pH ≈ 12-12.5
→ denaturate every thing
in cell

← خطوة ١ : plasmid كوي E-coli
← خطوة ٢ : Harvesting
← خطوة ٣ : TE
← خطوة ٤ : cell lysis

العليه بين خطوة ٤ و ٥ لازم تكون سريعه جداً .

← خطوة ٥ : اصفر ←لونه neutralization Solution
(renaturation)

1) $pH_{12} + pH_{5} \rightarrow pH_{7.5}$

$\boxed{1}$ neutralize pH

$\boxed{2}$ renaturation plasmid

2) SDS → PDS

$CH_3CH_2(11)SO_4\,Na^+ + K^+ \rightarrow CH_3CH_2(11)SO_4K^+ + Na^+$

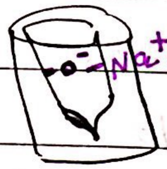
Soluble — insoluble

← خطوة ٧ : supernatant → plasmid + Na^+ + H_2O

pellet → cell debris + PDS

ننقله لـ spin column كاله

← خطوة ٨ :



← باقي الخطوات :

* washing buffer → ERB
→ CWC → main component is absolute ethanol

وكل مرة بنكب الـ (flow through) وبنرجع على نفس الـ collecting

* N.F.W → deionized H_2O

عشان تقدر تسحب الـ Na ions وتكسر الـ cationic bridge

وعشان هيك احنا بيصير استخدم الـ TE بدل N.F.W lab 5

* Agarose gel electrophoresis

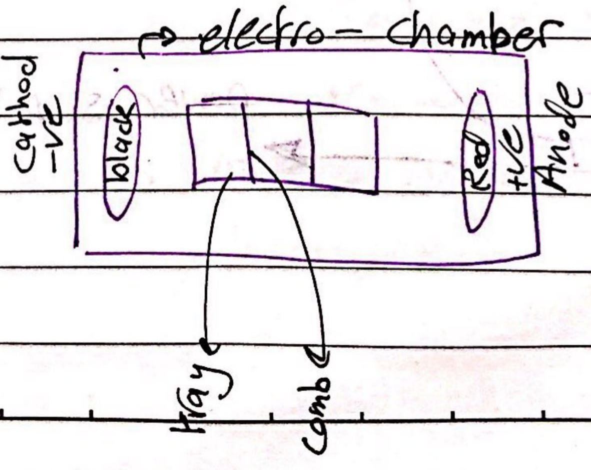
* Electrophoresis → technique used to separate and purify macromolecules (proteins, nucleic acids) that differ in size, charge or conformation.

* DNA molecules migrate in an electrical field and prossesses negative charge towards positively charge (anode)

− ⟶ +

* purposes for Agarose Gel electrophoresis:

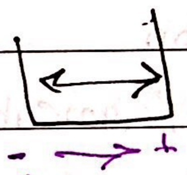
1) Analysis of DNA size & conformation
2) Separation and extraction of DNA fragments
3) DNA Quantitation
4) DNA degradation
5) Sizing of nucleic Acid:
   - DNA fragments for Southern transfer analysis
   - RNA molecules for North transfer analysis
   - Analytical separation of PCR products.
6) Detectio of mutations or variations.



Gel electro-

Nucleic acids
(Agarose)
هاد شائع استعماله اكثر
لانه هداك أصعب

Horizantal gel electro-

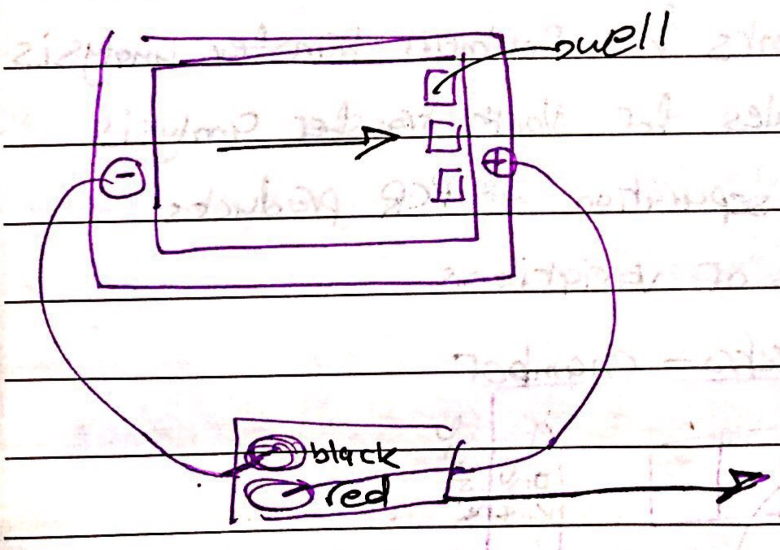


1D → size

protiens
(polyacrylamid)

Vertical gel

2D → size & charge



يكون مركز عشر مرات

10X

- Agarose gel :

1g agarose → 100 ml TBE buffer

1.5 g → 150 ml

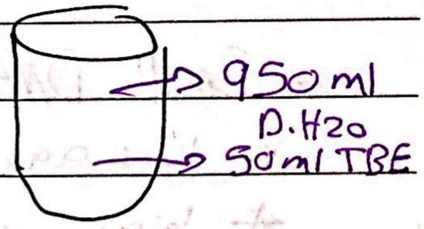
0.5 g → 50 ml

رنا بجرده

$C_1 V_1 = C_2 V_2$

$10X \; V_1 = 0.5X \; 1000 \, ml$

$V_1 = 50 \, ml$



then

dissolving in microwave

حتى يصير dissolve للagarose في الـ TBE

then

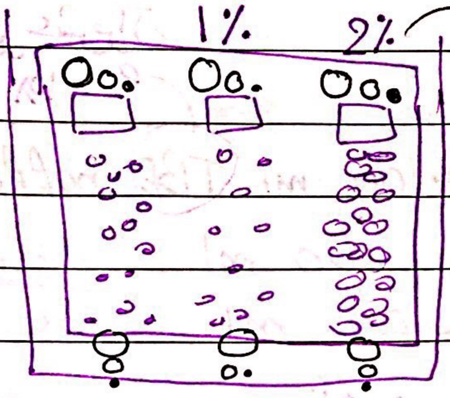
add DNA Stain (Ethidium bromide)

100 ml → 3 μl

50 ml → 2 μl

150 ml → 4 μl

* agarose → porous materia derived from ~~seaweed~~ Seaweed



* Small DNA fragment ~~faste~~ faster than large.
* low agarose → high resolution for large frag-
* high " → " " for small frag-

water+ions

* Function of buffers:
(carry applied current / Maintain pH / Determine electric charge on solute / inactive nucleases enzyme)

* TBE buffer: Tris-borate -EDTA
base (pH) / boric acid

* Ethidium bromide : (DNA stain)
→ aromatic → have aromatic ~~Hydrogen~~ Nitrogen base pair

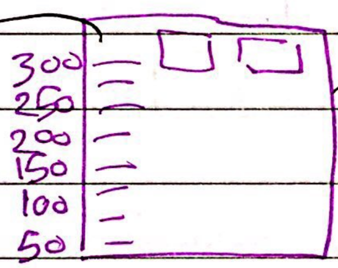
بدائله
→ acridine orange / Actinomycine D (expensive)

يدخل في تركيب الـ N.base و يعطيهم لون orange
لما نشوفهم تحت الـ UV

* Loading dye → denser materia (glycerol / sucrose): force DNA to fall down into well
→ Tracking dye: migrate in gel and allow visual monitoring or how far eletrophoresis proceeded
بضيف عينة ال DNA لتكون ثقيل ال dye فا
توصل للآخر يوقف التيار عشان ما نفقد ال DNA.

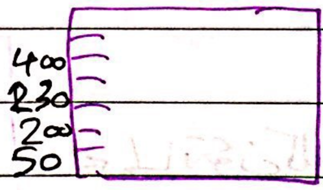
* DNA ladder:

ladder
50 bp
كل درجة
50



[redacted]

* DNA Marker



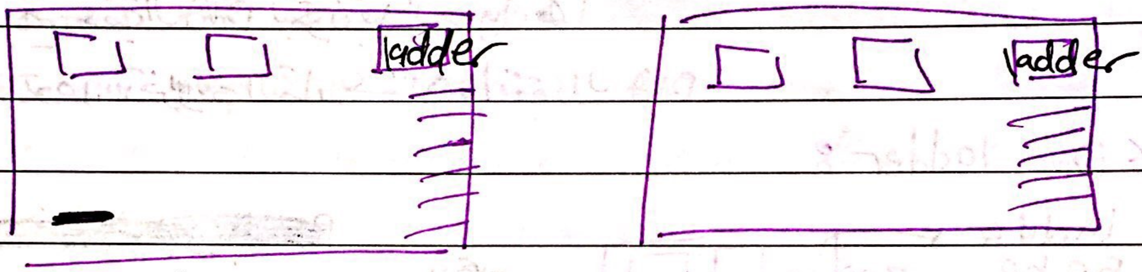
عدد غير ثابت
بين كل درجة

* intercalating agent ⟹ several hydrophobic molecules containing flat aromatic and fused heterocyclic rings can insert between stacked base pair of DNA

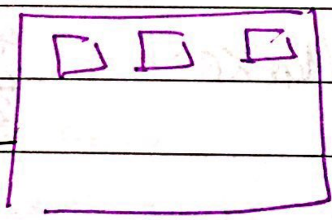
* Rate of migration of DNA determined by:
[1] DNA charge [2] strength of electrical field [3] DNA size
[4] DNA shape [5] state of hydration of DNA [6] Solution viscosity
[7] Gel pore size. [8] DNA stains.

علاقة عكسية: لما تزيد ال % بتزيد المقاومة
والمقاومة بتزيد درجة الحرارة
والحرارة بتذوب ال gel

Voltage Vol	Time	% Agarose
150	½ hr	1%
120	1:30 - 2 hr	2%
100	3 hr	3%



لما يكون ناقص Band واحد او كتير
والـ ladder موجود:
بكون الغلط في الـ isolation للعينة



لما تكون كلها فاضية حتى الـ ladder مو موجود
هون بنكون: ١) ما حطينا Ethidium
٢) وصلنا التيار غلط
٣) Too long run
اكتر من الوقت المطلوب

* متى بنحكي عنها كثر ال gel ← good quality DNA ?
when appears as a high molecular weight band with no or little evidence of smearing

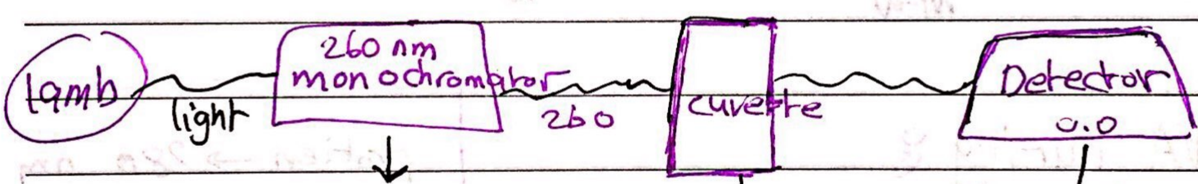
lab 6 (مهم جداً)   تكملة للاب السابق

*DNA Analaysis & Quintification

Methods:

1) Gel electrophoresis (Semi quintitave)
2) spectrophotometery
3) Nano drope ([illegible])
4) PCR

→ To determine concentration of substance in Solution by measuring light absorbed by solution at specific wave length



monochromator ↓ isolate wavelength and allow it to pass Through solution

cuvette ↓ sample placed in it

Detector ↓ Detect how much light absorbed / transmitted by / from sample

* Beer's lambert law:

$$C = \epsilon A \qquad C \propto A$$

C: conc.
ε: molar extinction Coefficent
A: light adsoption

سؤال للامتحان

* concentration of N.acids sample

$[dsDNA] = A_{260} \times (50\ \mu g/mL) \times$ dilution factor

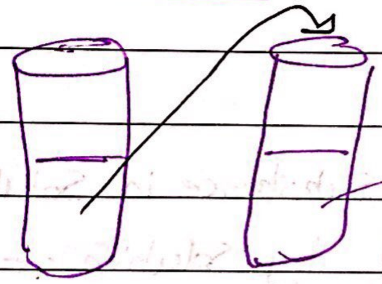
$[ssDNA] = A_{260} \times (33\ \mu g/mL) \times df$

$[ssRNA] = A_{260} \times (40\ \mu g/mL) \times df$

حفظ

Dilution

1 µl



DNA + 100 µl TE

99 µl + TE or NFW

spectro → A 260 nm → 0.128

$D = \frac{1 \times 1}{100} = 10^{-2}$

$D.F = \frac{1}{10^{-2}} = 10^{2}$

* DNA Purity :

$\hookrightarrow = \frac{A_{260}}{A_{280}}$

→ if [1.8 - 2] ∴ pure

Cont. واذا كانت اكثر او اقل بيكون

>2 ← RNA ← اما ب

<1.8 ← protien ←

protien → 280 nm

tyrosine, tryptophan } aromatic

* RNA Purity :

$\hookrightarrow = \frac{A_{260}}{A_{280}}$

→ [2.2 - 2.3] pure

DNA → > 2.3

protien → < 2.2

* ليش فضلنا ال gel عال spectro ؟

لانو العينة لما يكون صار إلها degraded حتى لو كانت pure والها

conc. عالي حارح ~~تبين~~ تبين الا بال gel ومش رح تزبط

بال PCR

* Nanodrop :
- * Small sample size (1-2 μl) only
- * no sample dilution
- * Fast and easy to use
- * Stores data on computer
- * converts OD to molarity.

* Using: 1) Turn on PC

2) open Nanodrop program

3) select DNA from menu

1- left arm 2- place 1μl water on pedestal to initialize instrment 3- blank instrument using water or elution buffer 4- place 1-2 μl of DNA sample on pedestal 5- click (Measure) on screen 6) recover sample or clean pedestal using wipe